File permissions in Linux

This is for a fictional scenario given as an activity for the *Tools of the Trade: Linux and SQL Course* in Coursera.

Scenario:

You are a security professional at a large organization. You mainly work with their research team. Part of your job is to ensure users on this team are authorized with the appropriate permissions. This helps keep the system secure.

Your task is to examine existing permissions on the file system. You'll need to determine if the permissions match the authorization that should be given. If they do not match, you'll need to modify the permissions to authorize the appropriate users and remove any unauthorized access.

File Permissions Document: Current file permissions

Project description

The research team's files needed their permissions checked so that only the right people have access. I used Linux commands to list the current permissions, find files and directories that were too open, and update them with `chmod` so each user and group has only the access they need.

Check file and directory details

I used the command `ls -la`
The `-l` option shows a long listing with permissions, owner, group, size, and date. The `-a` option also shows hidden files, which start with a dot, such as `.project_x.txt`.

The output is:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 ..
-rw--w---- 1 researcher2 research_team   33 Nov  6 23:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  6 23:24 drafts
-rw-rw-rw- 1 researcher2 research_team   33 Nov  6 23:24 project_k.txt
-rw-r----- 1 researcher2 research_team   33 Nov  6 23:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_t.txt
```

Describe the permissions string

Using the permissions for the project_m.txt as an example:

```
-  rw-  r--  ---
|  |    |    |
|  |    |    +-- others: no permissions
|  |    +------- group: read only
|  +------------ user: read and write
+--------------- file type ("-" is a file, "d" is a directory)
```

The string has 10 characters. The first shows the type. The next nine are three sets of three: user, group, and others. In each set, r is read, w is write, x is execute, and - means that permission is off.

Change file permissions

The organization does not allow others to have write access. In the output above, **project_k.txt** is **-rw-rw-rw-**, so others can write to it. It is the only file with this problem.

I used the command **chmod o-w project_k.txt**
o means others, - removes a permission, and w is write.

Updated permissions after using **ls -la**:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 ..
-rw--w---- 1 researcher2 research_team   33 Nov  6 23:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  6 23:24 drafts
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_k.txt
-rw-r----- 1 researcher2 research_team   33 Nov  6 23:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_t.txt
```

The **project_k.txt** line changed from **-rw-rw-rw-** to **-rw-rw-r--**.

Change file permissions on a hidden file

.project_x.txt is archived, which is why it is hidden. No one should be able to write to it, but the user and group should be able to read it.
I used the command **chmod u-w,g-w,g+r .project_x.txt**
This removes write from the user, removes write from the group, and adds read for the group.

Updated permissions after using **ls -la**:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 ..
-r--r----- 1 researcher2 research_team   33 Nov  6 23:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Nov  6 23:24 drafts
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_k.txt
-rw-r----- 1 researcher2 research_team   33 Nov  6 23:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_t.txt
```

The **.project_x.txt** line changed from **-rw--w----** to **-r--r-----**.

Change directory permissions

Only researcher2 should access the drafts directory. The group currently has execute permission (drwx--x---), which lets group members enter the directory.

I used the command: **chmod g-x drafts**
g means group, and x is execute. On a directory, execute means the person is allowed to enter it. Removing it stops the group from entering drafts.

Updated permissions after using **ls -la**:

```
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  6 23:24 ..
-r--r----- 1 researcher2 research_team   33 Nov  6 23:24 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Nov  6 23:24 drafts
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_k.txt
-rw-r----- 1 researcher2 research_team   33 Nov  6 23:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   33 Nov  6 23:24 project_t.txt
```

The **drafts** line changed from **drwx--x---** to **drwx------**

Summary

I reviewed the permissions in the projects directory with **ls -la**, found a file that others could write to, and removed that access. I then made the archived hidden file read-only for the user and group, and limited the **drafts** directory to its owner. These changes follow the organization's rules and lower the risk of unauthorized changes.